

Diverse divers



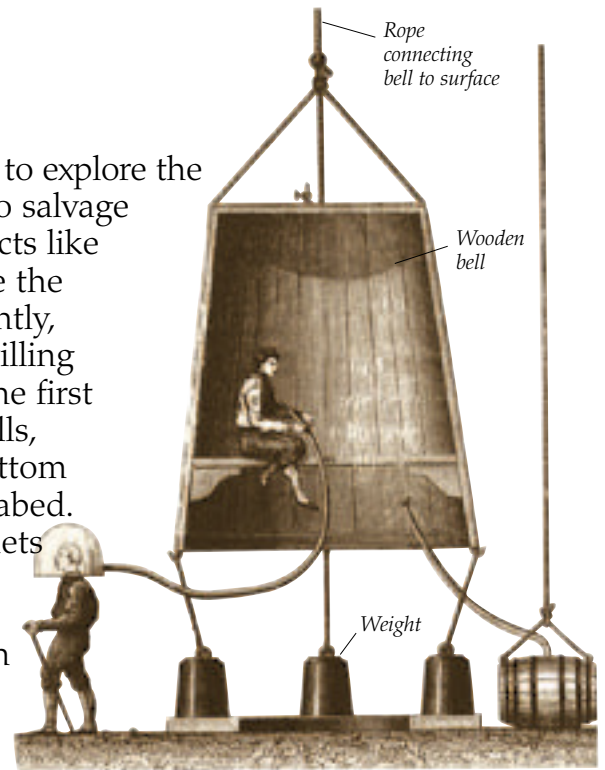
Umbilical
supplies
air and
electricity
for light

Weight
belt

UNDERWATER WORKER

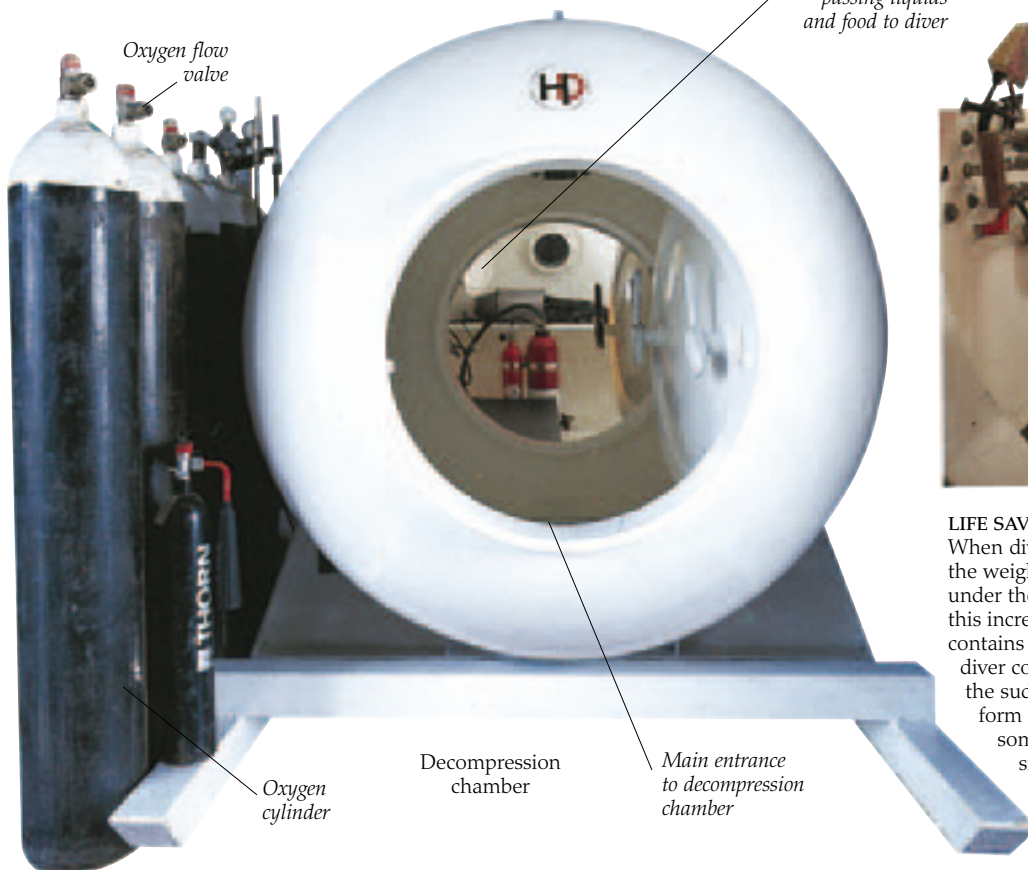
This diver, wearing a wetsuit for warmth, gets air into the helmet via a line linked to the surface. A harness goes around the diver's middle to carry tools. Flexible boots help the diver clamber around beneath an oil rig.

PEOPLE HAVE ALWAYS WANTED to explore the sea, to look for sunken treasure, to salvage wrecks, to bring up marine products like pearls and sponges, or to examine the beautiful underwater world. Recently, underwater oil exploration and drilling have also required divers' skills. The first diving equipment were simple bells, containing air and open at the bottom so the diver could work on the seabed. Later, diving suits with hard helmets were invented to enable divers to go deeper and stay down longer, with air pumped continually down a line from the surface. In the 1940s, the modern aqualung or SCUBA (Self-Contained Underwater Breathing Apparatus) was invented. Divers could carry their own supply of compressed air in tanks on their backs.



EARLY DIVING BELL

In 1690, Edmund Halley invented a diving bell, allowing a diver to be resupplied with barrels of air lowered from the surface. The bell, open at the bottom, was anchored to the seabed by heavy weights. A leather tube connected the lead-lined air barrel to the wooden bell. Several divers at a time could work from the bell which was used at depths of 60 ft (18 m).



"Medical lock" for
passing liquids
and food to diver

Joint pains indicate
decompression
sickness



LIFE SAVER

When diving, the pressure on the body increases due to the weight of water above the diver. The air is supplied under the same pressure so the diver can breathe. With this increased pressure, the nitrogen in the air supply (air contains 80 percent nitrogen) passes into the blood. If a diver comes up too quickly after a long or deep dive, the sudden release of pressure can cause nitrogen to form bubbles in the blood and tissues. This painful, sometimes fatal condition is called decompression sickness (the bends). The ailing diver is treated in a decompression chamber. The pressure is raised to the level required to move bubbles out through the lungs, and then is slowly reduced to normal pressure at the surface.